3. System Fundamentals

Before exploring the specific features of various modules related to the interaction concept in this research (as indicated in the Chapters 4, 5, 6, and 7), it is necessary to provide an overview of system architecture and test environments. This introduction is crucial to understanding the rationale and hypotheses behind the experiments and theories presented in these modules. Advanced systems show unique designs and functions, as discussed in Chapter 1. In the context of human interactions with vehicles, seemingly small characteristics such as the size and shape of the vehicle become important factors. The vehicles in question here are distinguished by the symmetrical front and rear design, lack of driver seats, lack of steering wheel. The absence of this driver leads to a decrease in pedestrian predictability [Guéguen 15] – due to the lack of regular driver gestures.

The Wizard of Oz technology [Detjen 20] is an attempt to simulate the functions of autonomous vehicles, focusing on simulation of driverless vehicles in experimental research. The observation indicated that pedestrians showed greater concern when they did not have a noticeable driver, which affected their confidence and decision-making processes when considering crossing the vehicle's path. This methodology allows a more authentic experimental setup. However, this method is harmful to the feasibility of driverless minibuses because there is no driver seat and thus enables the possibility of realistic experiments.

This section is aimed at investigating the design of the Autobus used in experimental validation of the concepts presented in this thesis. It includes a comprehensive description of the sensors installed on the vehicle and a detailed analysis of the safety systems integrated into the vehicle safety[1] chain. As already stated in the introduction (Chapter 1) prioritization of safety is crucial, especially due to pedestrians' involvement and the need to meet the certification standards for vehicle operations in public spaces. To explain the complexity associated with pedestrian navigation, this chapter provides a detailed study of the environment in which the vehicle has been tested. In addition, it presents a simulated version of this environment that was used for preliminary tests and partially to verify the feasibility and effectiveness of the proposed concept.